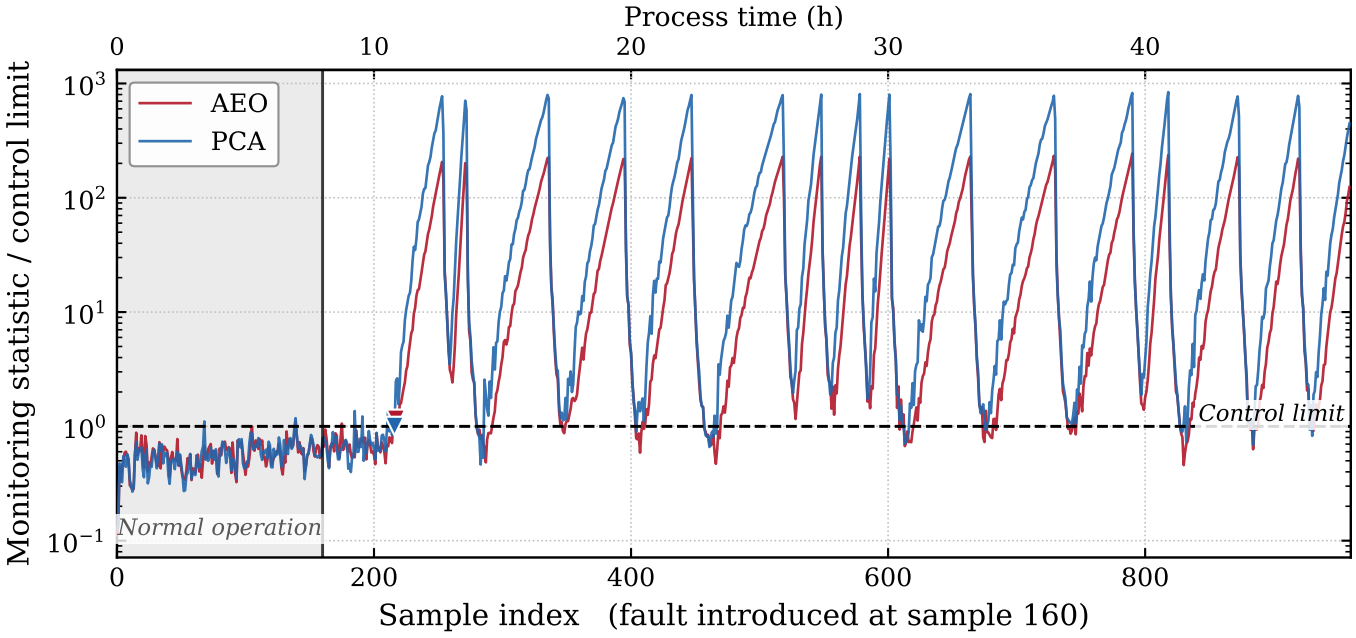
Fault Detection Performance for IDV(17)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	37.3 samples (112 min)	MDR 0.148	detected in 7/7 runs
PCA	delay	34.7 samples (104 min)	MDR 0.083	detected in 7/7 runs

PCA detects 2.6 samples (8 min) earlier; PCA missed-detection rate lower by 44%